

ANS Forth Test Suite, Organized by forth6809.asm

Section

Source: <https://forth-standard.org/standard/testsuite> (Annex F), fetched 2026-07-15.

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What's here

- `ttester.fs` — the test harness itself (ANS Forth Standard, Annex F, Section F.2.3), reorganized here into its own file rather than left bundled inline with anything else. Defines `T{ / -> / }T` and every supporting word (`ERROR` , `EMPTY-STACK` , the `HAS-FLOATING / HAS-FLOATING-STACK` detection, and the `X}T / R}T` -family closing words for mixed cell/float stack pictures). Load this first, before the prelude or any section file. Verified after transcription: `[IF]` and `[THEN]` counts match exactly (8/8), and `:` `/` ; counts match exactly (63/63) — no unbalanced conditional-compilation or colon definition introduced while extracting it from the fetched page.
- `00_test_prelude.fs` — shared setup every section depends on: the basic two's-complement assumptions (`BITSSET?`), the bound constants (`MAX-UINT`, `MAX-INT`, `MIN-INT`, `MID-UINT`, `MID-UINT+1`, `MSB`), the boolean constants (`0S/1S` via `<FALSE>/<TRUE>`), the floored-vs-symmetric division helpers (`IFFLOORED/IFSYM`), the string-compare helper (`S=`), and the shared memory buffers used by the `FILL/MOVE` tests (`FBUF/SBUF/SEEBUF`). Load this after `ttester.fs` , before any section file.
- `NN_name.tests.fs` — one file per forth6809.asm section (matching the exact same numbering and filenames as `forth6809_split/`), containing every ANS test block for the words that section implements.

153 test blocks placed across 17 section files (sections that implement no words with an official ANS test — e.g. section 5's inner-interpreter primitives like `LIT/BRANCH`, which have no directly-callable ANS name — have no file, since there was nothing to place there).

Load order — this matters, not just cosmetic

The original ANS suite is one continuous stream where later tests reuse words and

constants defined by earlier ones (this is explicit in the standard's own F.3 narrative: "these are included in the appropriate test"). Splitting it by forth6809.asm section preserves the *word groupings* but not the full original ordering — a few cross-section dependencies remain and must be respected:

- `09_outer_interpreter.tests.fs` (FIND) uses `GT1 / GT2`, which are defined by the `' / ['` tests in `13_compiling_words.tests.fs`. Load 13 before 9, even though forth6809.asm's own section numbering runs the other way.
- Several `20_numeric_output.tests.fs` tests use `MAX-BASE`, `#BITS-UD`, and `S=`, already provided by `00_test_prelude.fs`.
- `18_memory.tests.fs`'s `MOVE` test depends on the `FBUF / SEEBUF` state left behind by the immediately preceding `FILL` test in the same file — this dependency stayed intact since both are in the same section.

If you want a single flat run, the safe order is: `ttester.fs`, then `00_test_prelude.fs`, then sections in this sequence: 21, 17, 16, 14, 15, 18, 19, 20, 13, 9, 8, 12, 24, 10, 11, 26, 23 (roughly the original F.3 progression, adapted to where each word actually landed after the split).

Two real gaps this exercise surfaced — one still open, one since resolved

- **TRUE and FALSE were not implemented as dictionary words** when this test suite was first organized — checked directly against `forth6809.asm`'s dictionary section at the time: only the internal assembler constants `TRUEV / FALSEV` existed, never exposed as `CONSTANT TRUE / CONSTANT FALSE` a running program could call. This has since been resolved: both are now real dictionary words (`CONSTANT TRUE -1 / CONSTANT FALSE 0`), and their tests (`F.6.2.1485 FALSE`, `F.6.2.2298 TRUE`) are now included at the end of `26_abort_quit_headers.tests.fs`, alongside that section's other ROM-resident, hand-built words.
- Everything else in Core (F.6.1) and the Core Extension subset this system implements (F.6.2) mapped cleanly to exactly one section.

Not included

Double-Number, Facility, File-Access, Floating-Point, Memory-Allocation, and Search-Order word-set tests are all omitted — forth6809 implements none of those word sets (Search-Order's absence is documented explicitly in the ClaudeForth documentation, Section 4.5). Most of the Programming-Tools word set is also omitted (AHEAD, CS-PICK, CS-ROLL, N>R,

[THEN], etc. have no ANS test coverage relevant here since this system's own Tools word set — .S , WORDS , DUMP — are non-standard extensions with no official ANS test cases to begin with).